Kunwar Abhishek Singh

kasingh2694@gmail.com | +91-9720796764 | Linkedin.com/Kunwar

Google Scholar/Kunwar

Profile

I am a Joint Ph.D. graduate under the Melbourne India Postgraduate Academy [MIPA], jointly affiliated with Indian Institute of Technology Kharagpur, India and The University of Melbourne, Australia. My research focuses on advancing satellite-based remote sensing, machine learning, and data-driven modelling for monitoring and forecasting water quality in dynamic and ungauged river systems. I am particularly interested in integrating Earth observation, hyperspectral and multispectral remote sensing, artificial intelligence, and hydroclimatic information to develop scalable and transferable solutions for environmental monitoring and decision-making. **My broader research vision is to bridge environmental science with emerging computational and geospatial technologies to address complex challenges in water resources, ecosystem health, and environmental sustainability through interdisciplinary and international collaboration.**

Education

IIT Kharagpur, India and The University of Melbourne, Australia -(MIPA Candidate), Doctor of Philosophy — 2026

- CGPA: 9.4 (Coursework)
- **Broad Area of Research:** Water Quality Monitoring and Management

Indian Institute of Remote Sensing, Dehradun, India, Master of Technology in RS&GIS with specialization in Water Resources — 2020

- CGPA: 8.16

National Institute of Technology, Raipur, India, Bachelor of Technology in Civil Engineering — 2016

- CGPA: 8.66

Kendriya Vidyalaya, D.L.W, Varanasi, India, Intermediate — 2011

- Percentage: 92.6%

Kendriya Vidyalaya, D.L.W, Varanasi, India, Matriculation — 2009

- Percentage: 90%

Research Highlights and Publications

A robust forward-chain approach for ex-post and ex-ante seasonal river water-quality forecasting using remote sensing and hydroclimate forecast datasets. Published in *Journal of Water Process Engineering*, 93 (2026): 110920. [View Online] — 2026

Machine learning and remote sensing-based hierarchical framework for assessing non-optically observable riverine water quality parameters. Published in *Journal of Hazardous Materials Advances*, 22 (2026): 101111. [View Online] — 2026

Improving the Accuracy of Remotely Sensed TSS and Turbidity Using Quality Enhanced Water Reflectance by a Statistical Resampling Technique. Published in *International Journal of Applied Earth Observation and Geoinformation*, 142 (2025): 104681. [View Online] — 2025

Advanced Techniques for Mapping Water Quality in the Ganges. Published as a case study on the ARCH India website. [View Online] — 2024

Multi-mission Virtual Monitoring Station for Streamflow Monitoring. Published in *Remote Sensing Applications: Society and Environment*, 36 (2024) 101330. [View Online] 2024

Towards Robust Monitoring of Turbidity in the Ungauged Urban Reaches of Downstream Ganges River, India, using Sentinel-2 and Machine Learning Approaches *Presented in AGU Fall Meeting, Dec 2024, In-Person, Washington DC* [View Online] 2024

Suspended Particulate Matter Analysis of Pre and During Covid Lockdown Using Google Earth Engine Cloud Computing: A Case Study of Ukai Reservoir. Published in *Bulletin of Environmental Contamination and Toxicology*, 110:7 (2023). [View Online] 2023

Assessment of Water Quality Index Using Analytic Hierarchy Process (AHP) and GIS: A Case Study of a Struggling Asan River. Published in *International Journal of Environmental Analytical Chemistry*, 104:5 (2022), 1159-1171. [View Online] 2022

Positions of Responsibilities

Worked as the **Graduate Researcher representative**, IRTG-MIPA (University of Melbourne) Feb 2024 - Feb 2025

Worked as the head of public relation team of Entrepreneurship cell of NIT Raipur 2015 - 2016

Head coordinator and core member of the AAVARTAN (Techno-Management fest of NIT Raipur) 2014 and 2015

Organizer of the flagship event of the Entrepreneurship cell i.e. Entrepreneur Summit 2015

Awards and Achievements

Best oral presentation award on "Prediction of dissolved oxygen and electrical conductivity using remote sensing and data-driven models in an ungauged river" (Track: AI/ML in Water Quality Prediction and Control) at SCOPE–2025, IIT Kanpur - 2025

Melbourne Abroad Traveling Scholarship awarded by the University of Melbourne - 2024

Engineering and IT Conference Travel Grant awarded by the University of Melbourne - 2024

IITaV Inaugural **Young Researcher Award** awarded by IIT Alumni Association of Victoria - 2024

Melbourne Research Scholarship awarded by the University of Melbourne - 2022

Best paper award on "Analyzing long-term surface water change in Uttar Pradesh utilizing Google Earth Engine" in the VII ISEES International Conference on "Sustainable Energy and Environmental Challenges (SEEC-2022)"

Institute Assistantship (IIT Kharagpur), from the Ministry of Education (MoE), Government of India - 2021

Secured a position among **top 1.5% of total KVS students** who appeared in the All-India Senior School Certificate Examination 2011 conducted by the Central Board of Secondary Education (C.B.S.E)

Programming and Software Skills

Python

Google Earth Engine (GEE)

ArcGIS, QGIS, ENVI, SNAP

Personal Skills

Quick Learner

Leadership and Management

Problem-Solving Abilities

Creative Solutions

Hobbies and Interests

Dance (**freestyle, popping**)

Cricket (**all-rounder**)

Badminton (**competitive**)

Travelling (**road trips, mountain exploration**)

Storytelling (**crafting and sharing engaging narratives**)